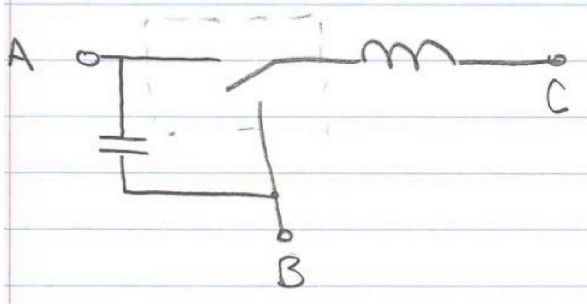


Lecture 6 — DC/DC Lecture 2

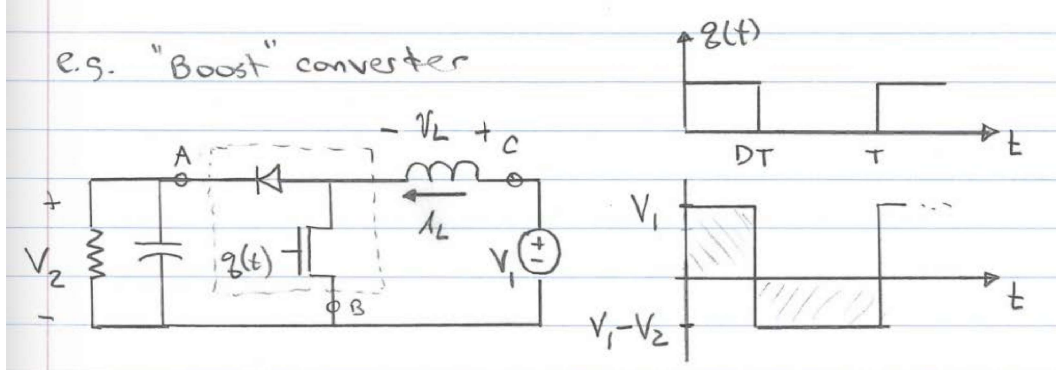
1 Switching

Last class we introduced the canonical switching cell:



We can construct the various types of converters from this.

E.g. "Boost" converter

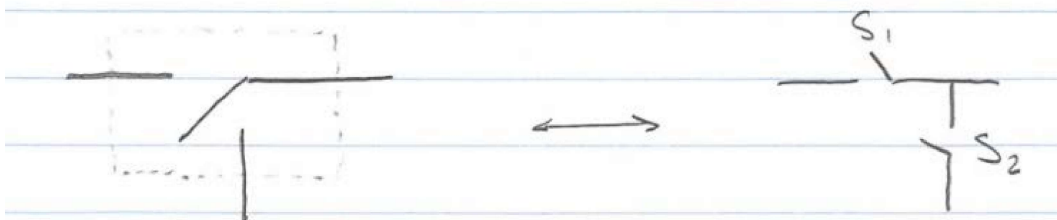


$$\langle V_L \rangle = DTV_1 + (1-D)T(V_1 - V_2) = 0 \text{ in P.S.S.}$$

$$\therefore V_1 = (1-D)V_2 \rightarrow \boxed{\frac{V_2}{V_1} = \frac{1}{1-D}}$$

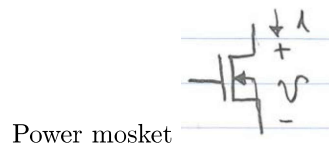
$$\text{by conservation of power } \boxed{\frac{I_2}{I_1} = 1-D}$$

The operational capabilities of a converter are heavily influenced by switch implementation



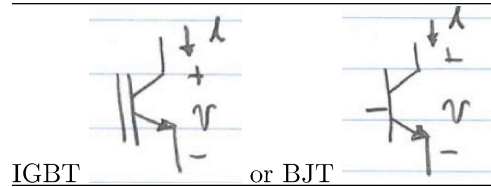
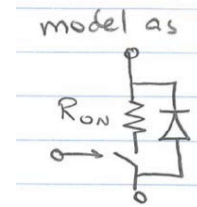
s_1, s_2 controlled so that inductor is never open-circuited with current through it.

Switch implementations:



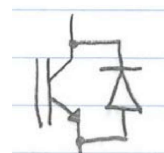
Can:
block +v
carry + or - i

Model as:

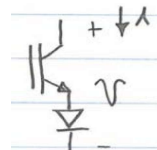


Similar, but cannot block → Most versions will reverse v or carry reverse i blow up!

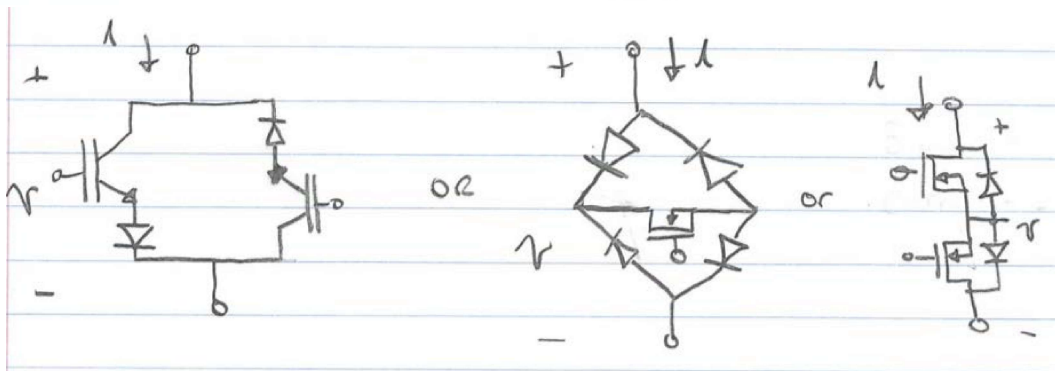
So:



block +v
carry +, - i



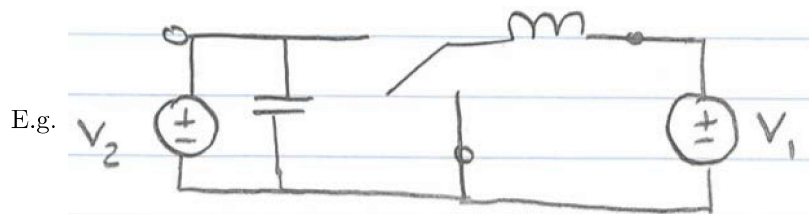
block +, = v
carry + i



Block +, - v
Carry +, - i

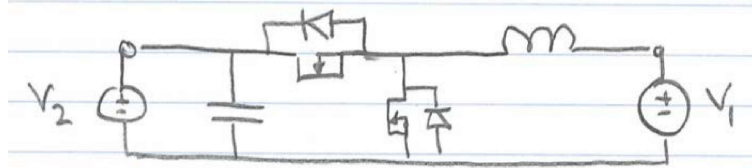
In analysing a converter and identifying power flow direction we need to know many things:

1. Switch implementation
2. External networks
3. Control



Can we tell power flow direction?
→ No

Adding switch implementation



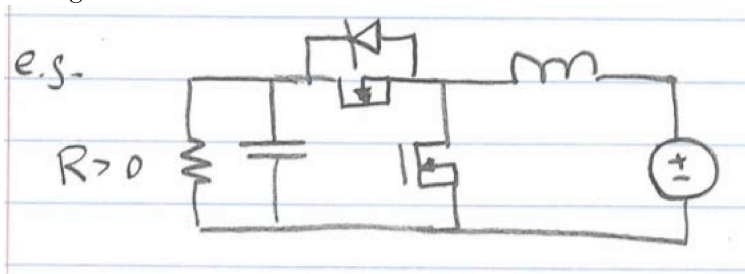
Note: the intronics/MIT converter is one example of a bidirectional unit of this type.

We still cannot tell

- Could do “buck” $L \rightarrow R$ or “boost” $R \rightarrow L$ operation (depends on control)
- We do know that v_1, v_2 must be > 0
- Also $v_2 \geq v_1$

External networks +/- or control may be sufficient.

E.g.

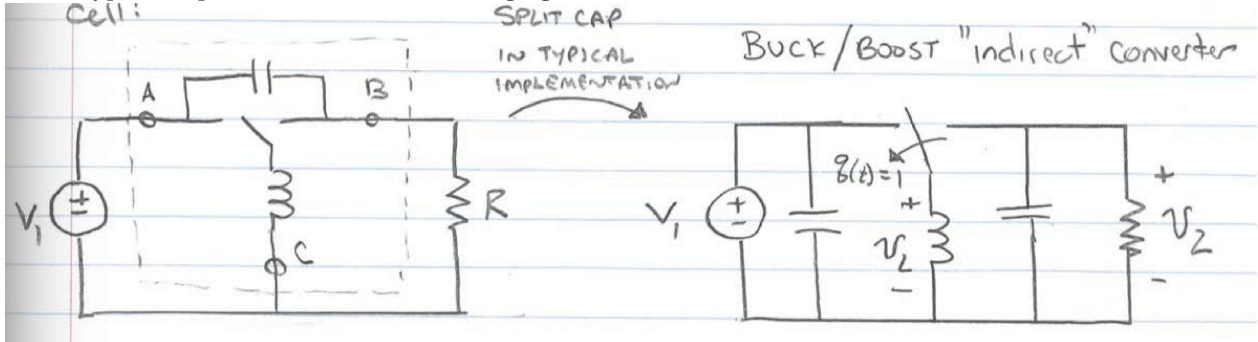


This must send power $R \rightarrow L$ in P.S.S. (but may transiently be bidirectional)

The buck, boost converters are direct converters

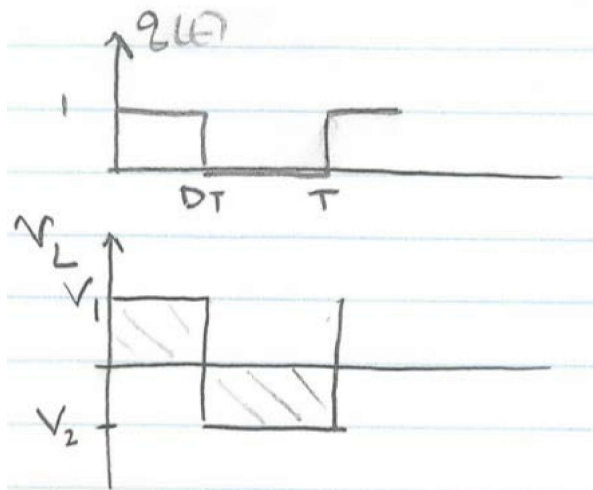
- energy transfer directly from one part to the other in one switch state.

Other types are possible. Consider rearranging the canonical cell:



Arrow on top: split cap in typical implementation.

Right text: Buck/boost “indirect” converter.



If L's, C's are big, in P.S.S. (small ripple)

$$\langle v_L \rangle = V_1 DT + V_2(1 - D)T = 0$$

$$\therefore v_2 = \frac{-D}{1-D} v_1$$

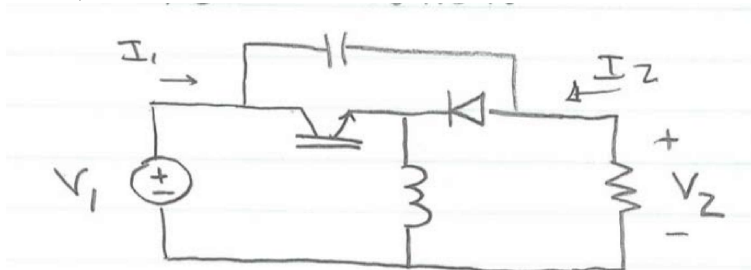
Assume $v_1 > 0$ in this case, $\therefore v_2 < 0$ so average 0!

$\rightarrow v_2$ has opposite sign to v_1
 $\rightarrow 0 < D < 0.5 \mid v_2 \mid < \mid v_1 \mid \mid 0.5 < D < 1 \mid v_2 \mid > \mid v_1 \mid$
 So as $0 < D < 1, -\infty < v_2 < 0$!

Explain:

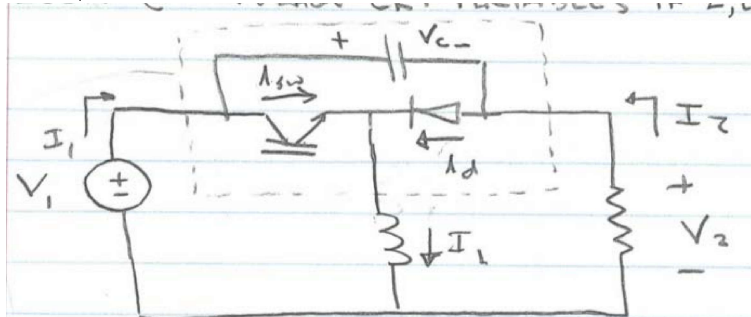
1. In 1st part of cycle, store incremental energy in L from v_1
2. 2nd part of cycle, transfer energy from L to $v_2 \rightarrow$ All energy is transferred via the inductor!
 $\rightarrow$ must have voltage inversion to get $\langle v_L \rangle = 0$

Switch implementation for $v_1 > 0, v_2 < 0$
 "Buck/boost" converter



If $v_1 > 0, v_2 < 0$
 $I_1 > 0, I_2 > 0$

Look @ average CKT variables if L, C very large (no ripple)
 "Buck/boost" converter



If L's, C's large
 $v_1(t) = V_1$
 $v_2(t) \approx V_2$
 $i_L(t) \approx I_L$

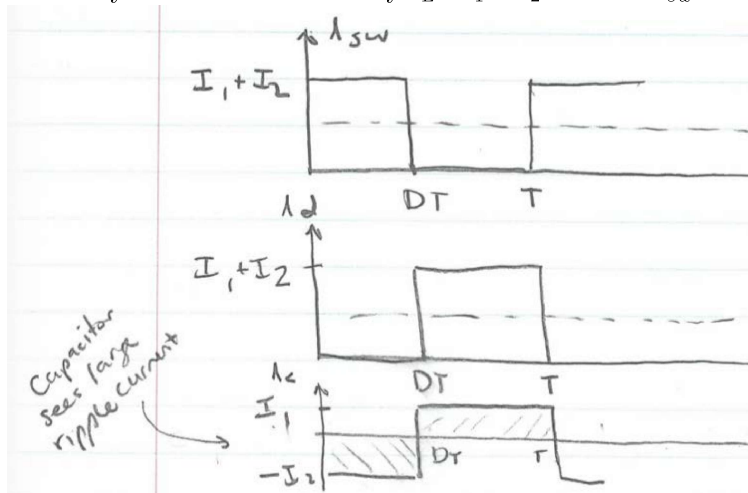
Average KCL into dotted box “supernode”:

$$I_L = I_1 + I_2 \text{ or } |I_L| = |I_1| + |I_2|$$

Average KVL around outer loop

$$V_c = V_1 - V_2 \text{ or } |V_c| = |V_1| + |V_2|$$

It maybe counterintuitive why $I_L \neq I_1 + I_2$: Look @ i_{sw}



$$i_{sw,peak} = I_1 + I_2$$

$$\langle i_{sw} \rangle = D(I_1 + I_2) = I_1$$

(since $\langle i_c \rangle = 0$)

$$i_{d,peak} = I_1 + I_2$$

$$\langle i_d \rangle = (1 - D)(I_1 + I_2) = I_2$$

(since $\langle i_c \rangle = 0$)

$$DI_2 = (1 - D)I_2$$

In general, for an **indirect converter**:

$$I_L = i_{z,peak} = i_{d,peak} = |I_1| + |I_2|$$

$$V_c = v_{q,peak} = v_{d,peak} = |V_1| + |V_2|$$

Show simulation of a buck/boost!

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6.622 Power Electronics
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